UNIT 9 — SOIL, AGRICULTURE AND ANIMAL HUSBANDRY

Comprehensive revision notes based on the supplied textbook pages, expanded with relevant concepts and examination facts.

PART A — AGRICULTURE

1. Agriculture and crop seasons

Agriculture is the practice of cultivating soil, growing crops and rearing useful plants and animals. Crops are grouped by growing season:

Rabi crops	Kharif crops
Grown mainly in winter. Examples: wheat, barley, oats, gram, peas, mustard and linseed.	Grown mainly during the monsoon/rainy season. Examples: rice, maize, bajra, groundnut, cotton, jowar and soybean.

2. Agricultural practices

Usual sequence: preparation of soil → sowing → adding manure/fertilisers → irrigation → weeding and crop protection → harvesting → threshing → winnowing → storage.

Preparation of soil

• **Ploughing/tilling:** loosening and turning soil with a plough, hoe or cultivator. It improves aeration, helps roots penetrate and mixes organic matter.

• **Levelling:** breaking clods and levelling the field. It helps distribute irrigation water more evenly and reduces soil/moisture loss.

Sowing

• **Broadcasting:** scattering seeds by hand; distribution may be uneven.

• **Seed drill:** places seeds in furrows at suitable depth and spacing, giving more uniform sowing and covering seeds with soil.

• **Transplantation:** seedlings are first raised in a nursery and then moved to the main field; used for paddy and some vegetables.

Weeding

• Weeds are unwanted plants growing with crops. They compete for light, water, nutrients and space, reducing crop yield.

• Weeds are removed manually, with tools such as a hoe, or by suitable herbicides/weedicides. Herbicides must be used carefully to avoid crop and environmental damage.

Irrigation

Irrigation is the artificial supply of water to crop fields at suitable intervals. Sources include wells, canals, rivers, tanks and reservoirs. Water is needed for photosynthesis, transport of minerals, germination and maintaining plant cells.

Method	Description / use
Drip	Water is supplied drop by drop near plant roots through pipes/emitters. Very water-efficient; useful in orchards, vegetables and water-scarce areas.
Sprinkler	Water is sprayed through nozzles like rainfall. Useful on uneven land; wind and evaporation may cause losses.
Furrow	Water flows through channels between rows of crops.
Basin	A level area around plants or a field section is filled with water; commonly used in paddy cultivation.

Drip vs sprinkler

• Drip delivers water directly to the root zone and usually reduces evaporation/runoff.

• Sprinklers distribute water over a wider area as droplets; wind drift can reduce efficiency.

Manures and fertilisers

Manure	Fertiliser
Organic material formed by decomposition of animal dung and plant residues. Adds humus and improves soil structure and water-holding capacity.	Material that supplies specific plant nutrients, often in concentrated and known amounts. Excess use can harm soil and water.

• **Manure examples:** farmyard manure, compost and green manure.

• **Nitrogenous fertilisers:** urea, ammonium sulphate, ammonium nitrate.

• **Phosphatic fertilisers:** superphosphate, ammonium phosphate.

• **Potassic fertilisers:** potassium chloride, potassium sulphate.

• Nitrogen is needed for proteins and chlorophyll; phosphorus supports roots and energy transfer; potassium supports enzyme activity and general plant health.

Crop protection

• **Pesticides** control pests such as insects and rodents. Use only appropriate products, in correct doses and with safety precautions.

• **Herbicides/weedicides** control weeds. Textbook examples include 2,4-D and 2,4,5-T; note that 2,4,5-T is restricted/banned in many places due to health and environmental concerns.

• **Fungicides** control fungal diseases. Examples include sulphur preparations and Bordeaux mixture (copper sulphate + lime).

• Excessive or careless pesticide use can harm humans, pollinators, birds, fish and other organisms. Integrated pest management combines monitoring, biological control and careful chemical use.

Harvesting, threshing, winnowing and storage

Practice	Meaning
Harvesting	Cutting and collecting mature crops; sickles and combine harvesters may be used.
Threshing	Separating grains from harvested stalks and husks.
Winnowing	Separating lighter chaff from heavier grains using moving air/wind.
Storage	Drying grains and storing them in clean, dry, pest-protected bins, granaries or containers.

• A combine harvester performs harvesting and threshing together. Proper drying prevents mould and reduces insect damage during storage.

3. High-yielding varieties (HYVs)

HYVs are crop varieties selected or developed for higher yield under suitable conditions. Yield still depends on water, nutrients, climate and management.

Crop	Textbook examples
Maize	Ambar, Ganga-5, Ganga-101, Jawahar
Rice	T-141, Jagannath, IR-8, Padma, Jaya
Wheat	Sarbati, Heera-Moti, Sonara, Kalyan Sona, V.P. 301

PART B — SOIL

4. Soil: formation and composition

Soil is the upper loose layer of Earth's crust in which plants grow. It forms slowly through weathering of rocks and the addition/decomposition of organic matter.

• Soil contains mineral particles (sand, silt and clay), humus, water, air and living organisms.

• **Humus** is dark organic matter formed by decomposition of dead plants and animals. It improves fertility and water-holding capacity.

• Soil texture depends mainly on the proportions of sand, silt and clay; soil structure describes how particles are grouped into aggregates.

5. Soil profile / layers

Layer	Main features
Topsoil (A-horizon)	Upper, usually dark layer; rich in humus, roots and organisms; generally fertile and biologically active.
Subsoil (B-horizon)	Below topsoil; often accumulates clay and minerals washed down from above; less humus and more compact.
Weathered parent material (C-horizon)	Partly weathered rock fragments and mineral material; little organic matter.
Bedrock	Hard underlying rock from which soil-forming material ultimately develops.

6. Types of soil by texture

Type	Properties
Sandy	Coarse particles and large pore spaces; good drainage and aeration but low water/nutrient retention. Often needs frequent irrigation and organic matter.
Clayey	Very fine particles; holds much water and nutrients but drains and aerates poorly. Sticky when wet and may become hard when dry.
Loamy	Balanced mixture of sand, silt and clay with humus. Usually has good drainage, aeration and water/nutrient retention; suitable for many crops.

Note: standard soil-science particle-size limits differ from simplified textbook descriptions. Remember the comparison: sand is coarse, silt intermediate and clay finest.

7. Soil erosion and conservation

Soil erosion is removal of topsoil by wind or flowing water. Topsoil is valuable because it contains much of the humus and biological activity.

• Causes: deforestation, overgrazing, removal of vegetation, heavy rainfall, floods, strong winds and improper farming on slopes.

• Effects: loss of fertility, lower crop yield, gullies, sedimentation of rivers/reservoirs and increased flood risk.

Conservation methods

• Plant trees and grasses; maintain vegetation cover and shelter belts.

• **Contour farming:** plough across a slope along contour lines to slow runoff.

• **Terracing:** create step-like fields on hillsides to reduce the speed of flowing water.

• Construct bunds, check dams and embankments to slow runoff and trap soil.

• Avoid overgrazing and deforestation; use crop rotation, cover crops and mulching.

• Add suitable manure/compost to maintain soil organic matter.

• Chipko Movement involved communities protecting trees; Van Mahotsav is an annual tree-planting initiative in India.

8. Soil pollution

Soil pollution is contamination of soil by harmful substances or waste that reduces soil quality and can harm crops, organisms and people.

• Sources: excessive agrochemicals, industrial effluents, sewage, garbage, plastics, contaminated irrigation water and hazardous waste.

• Excess irrigation in poorly drained areas can cause waterlogging and salinity; evaporation leaves salts behind and reduces crop productivity.

• Prevention: judicious agrochemical use, treatment of industrial waste and sewage, safe garbage disposal, avoiding open defecation and preventing polluted water from entering fields.

PART C — ANIMAL HUSBANDRY AND RELATED ACTIVITIES

9. Animal husbandry

Animal husbandry is the scientific care, breeding, feeding and management of domesticated animals for products and services such as milk, meat, eggs, wool, manure and transport.

• Good practices include suitable housing, balanced feed, clean water, hygiene, vaccination, disease control and selective breeding.

10. Cattle rearing

Cattle are reared for milk, draught power and manure. Cows and buffaloes are important dairy animals.

Cows	Buffaloes
Jersey, Holstein, Sahiwal, Sindhi, Lal Kandhari, Devani and Khillari.	Murrah, Surti, Nili, Carabao, Nagpuri and Bhadawari.

Cattle feed

• **Roughage:** bulky, fibrous feed such as cereal straw and green fodder; high in fibre and generally lower in concentrated nutrients.

• **Concentrates:** nutrient-rich feed such as oil cakes, grains and bran; provide more energy/protein per unit mass.

• A balanced ration also needs clean water, minerals and vitamins. Cottonseed oil cake and coarse grains may be used in feed mixtures.

Cattle diseases

• Foot-and-mouth disease is a contagious viral disease of cloven-hoofed animals. Anthrax is a serious bacterial disease. Foot rot affects feet/hooves.

• Prevention includes vaccination where recommended, clean housing, isolation of sick animals and veterinary care.

11. Poultry

Poultry farming is rearing domestic birds, especially chickens, for eggs and meat.

Type	Purpose and examples
Layers	Birds reared mainly for eggs. White Leghorn is a well-known layer breed.
Broilers	Birds reared mainly for meat; selected for rapid growth and efficient feed conversion. White Plymouth Rock and other meat strains are examples.

• Unfertilised eggs are commonly sold for eating. Fertilised eggs can develop into chicks when incubated under suitable conditions.

• Poultry require balanced feed, clean water, ventilation, hygiene and disease prevention.

12. Apiculture

Apiculture is rearing and management of honey bees for honey, beeswax and pollination. A place where colonies are kept is an **apiary**.

• A colony generally contains one queen, many workers and, during the breeding season, drones (males).

• **Queen:** reproductive female that lays eggs. **Workers:** sterile females that collect nectar and pollen, feed larvae, build/maintain comb and protect the colony. **Drones:** males whose main role is mating with a queen.

• Bees convert nectar into honey and store it in wax comb cells. Worker bees produce beeswax.

• Honey consists mainly of sugars (especially glucose and fructose), water and small amounts of other substances. Bees are important pollinators and can improve fruit and seed production.

13. Sericulture

Sericulture is rearing silkworms for silk. The common commercial silkworm is the larva of the silk moth, *Bombyx mori*.

• Silkworms feed mainly on mulberry leaves. The female moth lays eggs; eggs hatch into larvae/caterpillars, which spin a silk cocoon, form a pupa and finally emerge as adult moths.

• **Life cycle:** egg → larva/caterpillar → pupa inside cocoon → adult moth. Silk fibre is mainly a protein fibre obtained from the cocoon.

14. Pisciculture

Pisciculture is scientific breeding, rearing and harvesting of fish; it is a branch of aquaculture.

• Fish provide high-quality protein and nutrients including vitamins and minerals. Liver oils from some fish contain vitamins A and D.

• Fish farming may be carried out in ponds, tanks, cages or managed water bodies. Water quality, suitable feed, stocking density and disease control are important.

• Pearl oysters can be cultured to produce pearls; pearl culture is aquaculture but is not fish farming itself.

15. Important terms at a glance

Term	Meaning
Agriculture	Cultivation of crops and related farming activities.
Rabi / Kharif	Winter-season crops / monsoon-season crops.
Ploughing	Loosening and turning soil before sowing.
Seed drill	Tool placing seeds at suitable depth and spacing.
Transplantation	Moving nursery-raised seedlings to the main field.
Weeds	Unwanted plants growing among crops.
Irrigation	Artificial supply of water to crops.
Manure / Fertiliser	Organic decomposed matter / material supplying plant nutrients.
Threshing / Winnowing	Separating grain from stalk / separating chaff from grain using air.
Humus	Dark organic matter from decomposition.
Soil erosion	Removal of topsoil by wind or water.
Apiculture / Sericulture / Pisciculture	Rearing honey bees / silkworms / fish.
Animal husbandry	Scientific rearing and management of useful animals.

16. Quick Homi Bhabha facts

• Loamy soil is generally suitable for many crops because it balances drainage, aeration and water retention.

• Rice is commonly a Kharif crop and wheat commonly a Rabi crop in much of India.

• Drip irrigation is usually the most water-efficient of the common irrigation methods.

• Seed drill improves spacing and depth compared with broadcasting.

• Threshing separates grains from stalks; winnowing separates grains from lighter chaff.

• Topsoil is most vulnerable to erosion and is usually richest in humus.

• Layers are kept mainly for eggs; broilers are raised mainly for meat.

• Honey bees support pollination; silkworm cocoons provide silk; fish farming is pisciculture.

Study processes in sequence, compare similar terms carefully, and connect each practice with its purpose and practical use.